

Riverbend Community Solar Microgrid - Annual Performance Report 2024

Executive Summary

The Riverbend microgrid supplies electricity to 212 households and a primary school. In 2024 the system generated 1,480 megawatt-hours of electricity, which is 6 percent more than in 2023. The increase is mainly explained by the addition of 120 kilowatts of new panels on the school roof in March.

Battery storage allowed the community to cover 71 percent of its evening demand without drawing power from the regional grid. The average household electricity bill fell by 18 percent compared with the year before the microgrid was installed.

The main challenge of the year was an inverter failure in July that reduced output for eleven days.

Technical Performance

The solar array has a total installed capacity of 1.1 megawatts. The average capacity factor over the year was 15.4 percent. Output peaked in June and was lowest in December, when generation was about one quarter of the June value.

The lithium iron phosphate battery bank has a usable capacity of 2.4 megawatt-hours. Round-trip battery efficiency was measured at 89 percent. The battery management system recorded no thermal incidents during the year.

In July, a failed cooling fan caused one of the four central inverters to shut down. The inverter was replaced after eleven days, during which total output was reduced by roughly a quarter. The supplier covered the replacement under warranty.

Finances and Recommendations

Operating costs for 2024 were 96,000 euros, of which maintenance contracts accounted for 58 percent.

Revenue from selling surplus electricity to the regional grid was 41,000 euros.

Recommendations for 2025:

- Keep a spare inverter cooling fan and one spare inverter on site to shorten repair times.
- Expand battery capacity by 0.8 megawatt-hours to raise evening self-sufficiency above 80 percent.
- Introduce a time-of-use tariff that encourages households to run appliances around midday.
- Commission an independent audit of panel degradation, which the operator estimates at 0.5 percent per year.